

Task 2 — Linux Server Hardening

Ubuntu Server 26.04 LTS, hardened from a fresh install

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Target	Ubuntu Server 26.04 LTS (fresh VM, local/self-hosted)
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1. Objective

Take a fresh, default Ubuntu Server install and lock it down — the kind of basic hardening you'd want on any server before it goes anywhere near the internet. Covers user access, SSH, firewall, brute-force protection, patching, and password policy.

2. Tools Used

- UFW — firewall
- Fail2Ban — brute-force protection
- OpenSSH — remote access, hardened via config
- unattended-upgrades — automatic security patching
- libpam-pwquality — password policy enforcement

3. Methodology

Started from a clean Ubuntu Server 26.04 install with nothing customized yet. Worked through each hardening area one at a time — access control first (new user, no root SSH), then network-level protection (firewall, Fail2Ban), then long-term maintenance (auto-updates, password policy). After every change, ran a confirmation command to prove it actually took effect, not just that the command ran without error.

4. Steps Performed

New user, off root:

- `sudo adduser prajwal`
- `sudo usermod -s /bin/bash prajwal`
- Confirmed with: `groups prajwal`

Disabled root SSH login:

- Edited `/etc/ssh/sshd_config` → `PermitRootLogin no`
- `sudo systemctl restart ssh`
- Confirmed with: `sudo sshd -T | grep permitrootlogin`

Firewall:

- `sudo apt install ufw -y`
- `sudo ufw allow ssh`
- `sudo ufw enable`

- Confirmed with: `sudo ufw status verbose`

Brute-force protection:

- `sudo apt install fail2ban -y`
- `sudo systemctl enable --now fail2ban`
- Tested it by deliberately failing SSH logins from my Kali VM, then confirmed the IP got banned with: `sudo fail2ban-client status sshd`

Automatic security updates:

- `sudo apt install unattended-upgrades -y`
- `sudo dpkg-reconfigure --priority=low unattended-upgrades`
- Confirmed with: `cat /etc/apt/apt.conf.d/20auto-upgrades`

Password policy:

- `sudo apt install libpam-pwquality -y`
- Reviewed current settings with: `cat /etc/login.defs | grep PASS`

5. Screenshots

```

ubuntu@ubuntu:~$ sudo apt update && sudo apt upgrade -y
[sudo: authenticated] Password:
Hit:1 http://security.ubuntu.com/ubuntu resealite-security InRelease
Hit:2 http://ip.archive.ubuntu.com/ubuntu resealite InRelease
Hit:3 http://ip.archive.ubuntu.com/ubuntu resealite-updates InRelease
Hit:4 http://ip.archive.ubuntu.com/ubuntu resealite-backports InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  Use 'sudo apt autoremove' to remove them.

Upgrading:
  libapparmor1          linux-firmware-marvell-wireless  packagekit
  apport                linux-firmware-mediatek          python3-apport
  apport-core-dump-handler  libcrda2to                      python3-distupgrade
  base-files            libpackagekit-glib2-18          linux-firmware-misc             python3-problem-report
  bind9-dnsutils        libglib2.0-1.40                 linux-firmware-netronome       sg3-utils
  bind9-host            linux-firmware-amd-graphics     linux-firmware-oville-graphics  sg3-utils-udev
  bind9-libs            linux-firmware-amd-wildc         linux-firmware-qlogic          ubuntu-kernel-accessories
  cloud-init            linux-firmware-broadcom-wireless  linux-firmware-qualcomm-graphics  ubuntu-minimal
  head                 linux-firmware-intel-graphics    linux-firmware-qualcomm-misc     ubuntu-release-upgrader-core
  iproute2              linux-firmware-intel-wireless    linux-firmware-qualcomm-wireless  ubuntu-server
  iproute2              linux-firmware-marvell-prestera  linux-firmware-realtek          ubuntu-server-minimal
  iproute2              linux-firmware-marvell-prestera  notd-new-config                ubuntu-standard

Not upgrading yet due to phasing:
  grub-pc  grub-pc-bin  grub2-common  python3-software-properties  software-properties-common

Summary:
  Upgrading: 48, Installing: 0, Removing: 0, Not Upgrading: 5
  Download size: 713 MB
  Space needed: 1,138 MB / 6,566 MB available

0% [Working]
  
```

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File Edit View VM Tabs Help
Kali Linux x My Computer x Ubuntu 64-bit x

Hit:1 http://in.archive.ubuntu.com/ubuntu resolve-backendports InRelease
S3 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14 pollinate
Use 'sudo apt autoremove' to remove them.

Upgrading:
apparmor libapparmor1 linux-firmware-nvidia-wireless packagekit
apport libapt3 linux-firmware-nvidia-wireless python3-apport
apport-core-dump-handler libcrypt2 linux-firmware-nvidia-wireless python3-distupgrade
base-files libpackagekit-glib2-1b linux-firmware-nvidia-wireless python3-problem-report
bind9-host libqt5-qt5 linux-firmware-nvidia-wireless sg3-utils
bind9-libs linux-firmware-nvidia-wireless sg3-utils-udev
cloud-init linux-firmware-nvidia-wireless ubuntu-kernel-accessories
cloud-init-base linux-firmware-nvidia-wireless ubuntu-minimal
cloud-init-base linux-firmware-nvidia-wireless ubuntu-release-upgrader-core
fwupd linux-firmware-nvidia-wireless ubuntu-server
git1.2.2-packagekit-glib-1.0 linux-firmware-nvidia-wireless ubuntu-server-minimal
iproute2 linux-firmware-nvidia-wireless ubuntu-standard
Not upgrading yet due to phasing:
grub2-common grub2-common python3-software-properties software-properties-common

Summary:
Upgrading: 48, Installing: 0, Removing: 0, Not Upgrading: 5
Download size: 715 MB
Space needed: 1.198 KB / 6.506 MB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 motd-news-config all 14ubuntu1.1 [5,632 B]
Get:2 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 base-files amd64 14ubuntu1.1 [75,0 KB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 libcrypt2 amd64 1:12.0-2ubuntu1 [671 KB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 iproute2 amd64 5.19.0-1ubuntu1 [1,139 KB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 libapparmor1 amd64 5.0.2-ubuntu1 [51.3 KB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 ubuntu-minimal amd64 1:17.0-2 [8,594 B]
Get:7 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 apparmor amd64 5.0.2-ubuntu1 [309 KB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 bind9-host amd64 1:9.20.24-1ubuntu1 [49,0 KB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 bind9-dnssync amd64 1:9.20.24-1ubuntu1 [164 KB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu1 [1,388 KB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 ubuntu-release-upgrader-core all 1:26.1-0ubuntu1 [9,674 B]
Get:12 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 python3-distupgrade all 1:26.1-0ubuntu1 [102 KB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 python3-problem-report all 1:26.1-0ubuntu1 [17,4 KB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 python3-apport all 2.34.1-0ubuntu1 [96,4 KB]
Get:15 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu1 [9,674 B]
Get:16 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 sg3-utils amd64 1:1.48-1ubuntu1 [83,9 KB]
Get:17 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 sg3-utils-udev all 1:1.48-1ubuntu1 [597 KB]
Get:18 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 cloud-init all 26.1-0ubuntu1 [1,218 B]
Get:19 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 cloud-init-base all 26.1-0ubuntu1 [1,218 B]
Get:20 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 fwupd amd64 2.1.1-1ubuntu1 [12,7 MB]
Get:21 http://in.archive.ubuntu.com/ubuntu resolve-backendports/main amd64 git1.2.2-packagekit-glib-1.0 amd64 1:1.2.2-1ubuntu1 [12,7 MB]
195 kB/s 1h 52min 49s_

To return to your computer, press Ctrl+Alt.

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Kali Linux x My Computer x Ubuntu 64-bit x

fwupd-refresh.service is a disabled or a static unit not running, not starting it.
fwupd.service is a disabled or a static unit not running, not starting it.
Setting up sg3-utils (1.48-1ubuntu1.1).
Setting up cloud-init (26.1-0ubuntu1.1).
Setting up ubuntu-minimal (1:17.0-2).
Setting up sg3-utils-udev (1.48-1ubuntu1.1).
update-initramfs: defer update (trigger activated)
Setting up bind9-dnssync (1:9.20.24-1ubuntu1.1).
Setting up ubuntu-standard (1:17.0-2).
Setting up python3-distupgrade (1:26.1-0ubuntu1.1).
Setting up python3-problem-report (1:26.1-0ubuntu1.1).
Setting up python3-apport (2.34.1-0ubuntu1.1).
Setting up apport-core-dump-handler (2.34.1-0ubuntu1.1).
Setting up ubuntu-server (1:17.0-2).
Setting up ubuntu-server-minimal (1:17.0-2).
Processing triggers for plymouth (24.0-0ubuntu1.1) [120250831.433c171d-ubuntu].
Processing triggers for libc-bin (2.40-0ubuntu1.1).
Processing triggers for procps (3.2.0-1ubuntu1.1).
Processing triggers for system (259.5-0ubuntu1.1).
Processing triggers for man-db (2.13.1-1ubuntu1.1).
Processing triggers for dbus (1.16.2-0ubuntu1.1).
Processing triggers for dracut (10-11).
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Processing triggers for procps (2:4.0.4-0ubuntu1.1).
Processing triggers for install-info (7.2-0ubuntu1.1).
Processing triggers for plymouth-theme-ubuntu-text (24.0-0ubuntu1.1) [120250831.433c171d-ubuntu].
Processing triggers for dracut (10-11).
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Scanning processes...
Scanning candidates...
Scanning Linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/needrestart.d/systemd-manager

Service restarts being deferred:
/etc/needrestart/needrestart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ user manager: (sd-pam) [1501]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ubuntu:~$ sudo useradd prajwal
ubuntu@ubuntu:~$ sudo usermod -s /bin/bash prajwal
ubuntu@ubuntu:~$ groups prajwal
prajwal : prajwal sudo
ubuntu@ubuntu:~$ sudo _
```



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[Ubuntu 64-bit]

# ListenAddress.
#
# For changes to take effect, run:
# systemctl daemon-reload
# systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#KeyExchange default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

ubuntu@ubuntu:~$ sudo systemctl restart ssh
ubuntu@ubuntu:~$ sudo apt install ufw -y
ufw is already the newest version (0.36.2-9build1).
ufw set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14      pollinate
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 5
ubuntu@ubuntu:~$

To return to your computer, press Ctrl+Alt.

File Edit View VM Tabs Help
[Ubuntu 64-bit]

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
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  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14      pollinate
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 5
ubuntu@ubuntu:~$ sudo ufw allow ssh
Rules updated (0s)
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo apt install fail2ban -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14      pollinate
Use 'sudo apt autoremove' to remove them.

Installing:
  fail2ban

Installing dependencies:
  python3-pysnmpcore python3-pyinotify whois

Suggested packages:
  mailx monit fail2ban python-pyinotify-doc

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 5
  Download size: 510 kB
  Space needed: 2,620 kB / 6,504 MB available

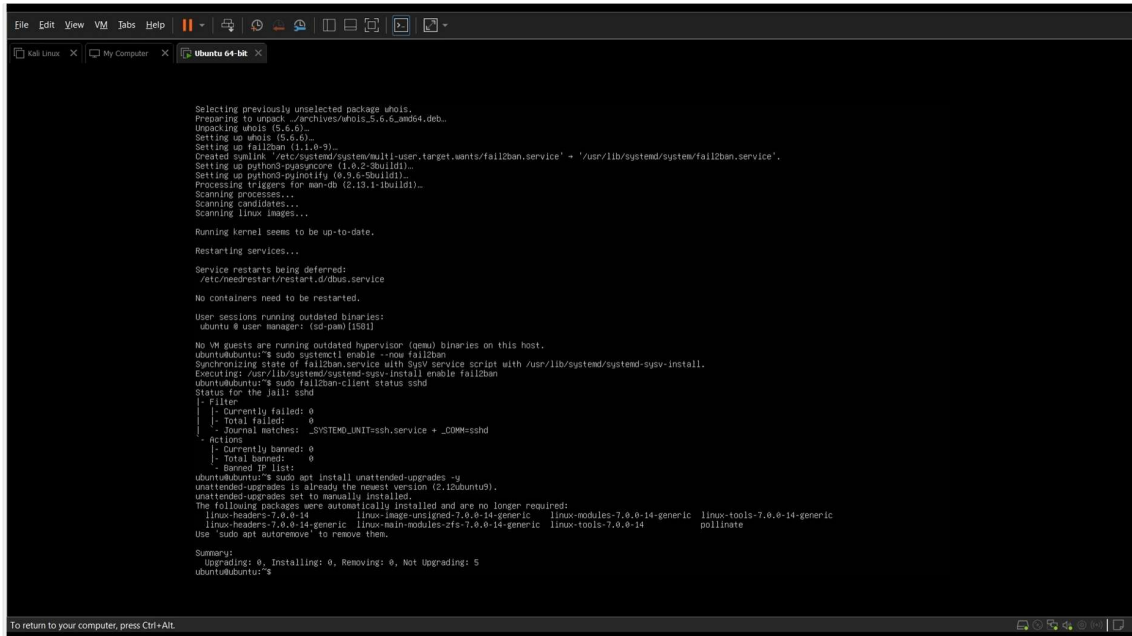
ok [working]_

To return to your computer, press Ctrl+Alt.
```



Task 5

Enable automatic updates



6. Findings

What the server looked like by default, and what changed:

Area	Before → After	Risk if left unfixed
User access	Only the install-time account existed, with full admin rights baked in from setup. Added a separate sudo user instead.	Single point of failure
Root SSH	Root login over SSH was left to OpenSSH's default (prohibit-password, commented out). Explicitly disabled it.	Common brute-force target
Firewall	No firewall active at all — every port was reachable by default.	Wide open attack surface
Brute-force protection	Nothing was watching failed login attempts.	Unlimited password guessing
Patching	Updates required manually running apt upgrade.	Delayed security patches
Password policy	Default Ubuntu settings only — no complexity enforcement.	Weak passwords allowed

7. Risk Analysis

None of these are exotic vulnerabilities — they're all just default settings that assume you'll harden them yourself, and most breaches of small/misconfigured servers happen exactly because nobody did. The firewall being off is the biggest one: it means every service on the box, not just SSH, was reachable from anywhere. Fail2Ban and the SSH change work together — even if someone finds the right username, they're stuck guessing passwords with no automation, and get temporarily blocked for trying too hard.

Auto-updates and password policy are more about long-term posture than an immediate hole — they reduce how long a known issue stays exploitable, and how easy a compromised account is to get into in the first place.

8. Recommendations

- Keep root SSH login disabled permanently — no legitimate reason to re-enable it
- Review UFW rules periodically as new services get added, so only what's needed stays open
- Consider SSH key-based login instead of passwords entirely, for a further step up

- Actually enforce the password quality rules (minimum length, complexity) rather than just having the module installed
- Set a reminder to check unattended-upgrades logs occasionally — automatic doesn't mean 'never look at it'

9. Conclusion

Took a completely default Ubuntu Server install and worked through the basics that any real deployment needs before going live — proper user separation, locked-down SSH, an active firewall, brute-force protection that I actually tested rather than just installed, automatic patching, and a password policy check. Each step got verified with its own confirmation command rather than just assuming it worked, which is really the habit this task was meant to build.